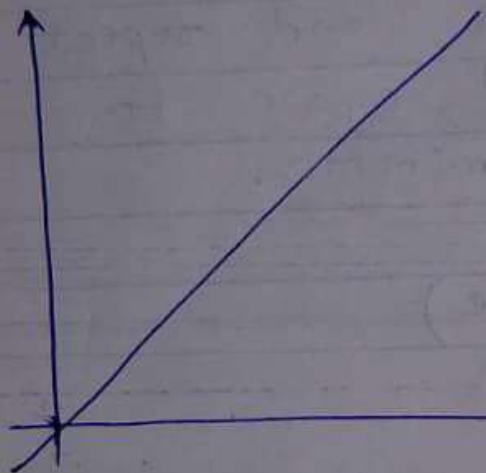


$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

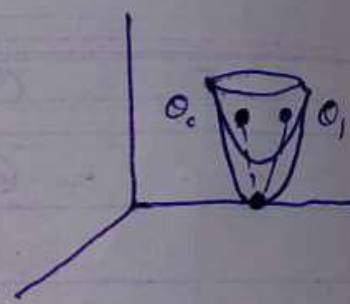
Final Conclusion

GRADIENT DESCENT



$$\theta_0 = 0$$

$\theta_1 = \text{Changeable}$



Convergence Algorithm

repeat until convergence

1

$$\theta_j = \theta_j - \alpha \left[\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) \right]$$

2

$j = 0 \text{ and } 1$

$n-1$

$$\frac{\partial}{\partial x} x^n = nx^{n-1}$$

$$\frac{\partial}{\partial x} (x)^2 = 2x$$

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_j} \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x)^{(i)} - y^{(i)})^2$$

$$h_{\theta}(x) = \theta_0 + \boxed{\theta_1 x} \rightarrow 0$$

if $j=0 \Rightarrow \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) =$

$$= \frac{\partial}{\partial \theta_0} \frac{1}{2m} \left[\sum_{i=1}^m (h_{\theta}(x)^{(i)} - y^{(i)})^2 \right]$$

$$= \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x)^{(i)} - y^{(i)}) \times 1$$

Note: Differentiate with θ_0 so θ_1 is constant
so, $\theta_1 x = 0$ and $\theta_0 = 1$

$j=1 \Rightarrow \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$

$$= \frac{\partial}{\partial \theta_1} \frac{1}{2m} \left[\sum_{i=1}^m ((\theta_0 + \theta_1 x) - y^{(i)})^2 \right]$$

$$= \frac{1}{m} \sum_{i=1}^m ((\theta_0 + \theta_1 x) - y^{(i)}) (x)$$

$$\frac{\partial}{\partial \theta_1} [\theta_0 + \theta_1 x] \Rightarrow x$$

Repeat until Convergence

$$\theta_0 = \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_0(x)^{(i)} - y^{(i)})$$

$$\theta_1 = \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m (h_0(x)^{(i)} - y^{(i)}) x^{(i)}$$